

Jie Kang

Applied Statistician

Quantitative genetics | Biomedical data science | Precision livestock farming

University of Sydney

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RESEARCH PROFILE

I am an applied statistician working across quantitative genetics, biomedical data science and precision livestock farming. My research develops scalable statistical, machine-learning and AI-enabled frameworks for complex biological and agricultural data, including genetic, omics and sensor-derived datasets. I translate high-dimensional data into decision-support tools for agricultural and biomedical research.

CURRENT APPOINTMENTS

Lecturer in Applied Statistics 2025–present
University of Sydney

- Develop and apply statistical and machine-learning methods for complex agricultural and biomedical research data.
- Advance collaborative research through the Charles Perkins Centre Data Science Hub, including study design, predictive modelling, reproducible analysis and decision-support workflows.
- Co-lead curriculum development in data-driven decision-making and translational data science.

Chief Technology Officer 2026–present
SureStock

- Lead the technical roadmap for livestock bioanalytics, with a focus on biomarker discovery and predictive modelling of immune fitness.
- Oversee the development of scalable decision-support tools to identify animals with divergent immune fitness and production potential.

EDUCATION

2018–2023 **PhD in Quantitative Genetics**, University of Otago.
2013–2017 **BSc(Hons) in Mathematics and Statistics**, University of Otago.

RESEARCH FUNDING AND COMMERCIALISATION

Awarded funding

Year(s)	Scheme / funder	Project and impact	Role	Value
2026–present	Australia’s Economic Accelerator Ignite	SureStock: biomarker-driven selection to identify feedlot-ready cattle.	Co-entrepreneur / technical lead	A\$489,710
2018–2023	Teagasc Walsh Scholarship	PhD research in quantitative genetics and genomic selection using reduced-representation sequencing data.	PhD scholar	EUR 22,000 p.a.

Major funded programmes contributed to

Year(s)	Programme / funder	Contribution	Programme scale
2021–2025	Informing New Zealand Beef, Beef + Lamb New Zealand / Ministry for Primary Industries	Led methodological development and delivery of multi-breed beef cattle genetic evaluations for New Zealand breeders.	NZ\$16.7m programme; NZ\$460m projected sector benefit
2025–present	Eliminating Facial Eczema Impacts, Beef + Lamb New Zealand	Developed workflows for spore-count integration, epidemiological mapping, risk reporting and animal-health decision support.	NZ\$20.75m programme
2025–present	Charles Perkins Centre Data Science Hub, University of Sydney	Contribute statistical modelling, machine learning and reproducible data science expertise to collaborative biomedical research.	A\$1.36m over three years; A\$189,000 Faculty of Science co-contribution

KEY RESEARCH CONTRIBUTIONS

Quantitative genetics. Developed and implemented statistical methods for genomic prediction, multi-breed genetic evaluation and breeding programmes, with applications across beef cattle, sheep, potato and perennial ryegrass.

Biomedical and omics data science. Applied statistical learning and machine-learning methods to high-dimensional biological data, including genomics, genotyping-by-sequencing, transcriptomics, proteomics and multi-modal biomarker datasets, with applications in biomedical risk prediction, genetic improvement and animal health.

Precision livestock farming. Contributed machine-learning, deep-learning and AI-enabled workflows for livestock phenotyping, animal-health surveillance, biological prediction and decision-support systems. Developed open-source software and reproducible analytical workflows for complex biological data.

PROFESSIONAL EXPERIENCE

Genetic Evaluation Specialist
Beef + Lamb New Zealand

2021–2025

- Led development and delivery of New Zealand’s multi-breed beef cattle genetic evaluation system.
- Developed workflows for national-scale genetic, phenotypic and animal-health datasets.
- Worked with researchers, industry partners and breeders to translate quantitative genetics methods into practical genetic evaluation tools.
- Established infrastructure supporting multimodal animal-health prediction and epidemiological reporting.

SELECTED RESEARCH OUTPUTS

1. **Kang, J.**, Hess, M. K., Dodds, K. G., Brauning, R., McEwan, J. C., Foote, B. J., ... & Hess, A. S. (2026). SimGBS: a rapid method for simulating large-scale genotyping-by-sequencing data. *BMC Bioinformatics*, 27(1), 12.
2. Zhao, X., Liao, Q., Wang, D., Meijering, E., Brosnahan, C., Keeling, S., Tian, R., Li, W., Sutherland, M., & **Kang, J.** (2026). Multimodal animal health monitoring in extensive livestock production systems. *Frontiers in Veterinary Science*, 13, 1832869.
3. Sun, E., Sun, X., Qin, W., Hao, S., Chen, Y., Sun, L., Jin, Y., Wang, N., Tian, R., Li, H., **Kang, J.**, Xiao, X., & Zhao, L. (2026). Effects of *Pediococcus pentosaceus* and tea polyphenols on biogenic amines, volatile compounds, and microbial community in fermented sausage. *Food Chemistry*, 148596.
4. Yue, C., **Kang, J.**, Liu, C., Han, B., Li, Z., & Li, W. Estimation of genetic parameters for wool traits in FGF5 gene-edited Chinese Merino sheep. *Frontiers in Animal Science*, 6, 1712295.
5. Ling, Q., Yu, H., Yue, C., **Kang, J.**, Maimaiti, A., Li, Z., ... & Li, W. (2025). A dual-branch multimodal deep learning framework for non-destructive evaluation of intramuscular fat in sheep. *Scientific Reports*.
6. Vexler, L., Konkolewska, A., Byrne, S., Ruttink, T., Leyva-Pérez, M. D. L. O., **Kang, J.**, ... & Milbourne, D. (2025). PotatoMASH is a cost-effective marker system for genomic prediction in potato based on short-read haplotypes. *Theoretical and Applied Genetics*, 138(11), 1–20.
7. **Kang, J.**, Garrick, D. J., Lee, M. A., Weik, F., & Archer, J. A. (2025). Across-breed analysis of maternal traits in New Zealand beef cattle. *Proceedings of the Association for the Advancement of Animal Breeding and Genetics*, 26, 272–275.
8. Cameron, K., Toldi, G., Owen, M., Kemper, J., Truax, O., Hughes, E., ... & **Kang, J.** (2024). Early Career Researchers succeeding under a changing research system. *New Zealand Science Review*, 80.

9. Wang, T., Griffin, J. D., Brenna, M., Fletcher, D., Zeng, J., Stirling, M., ... & **Kang, J.** (2024). Earthquake forecasting from paleoseismic records. *Nature Communications*, 15(1), 1944.
10. **Kang, J.**, Weik, F., Sanderson, N., Robertson, D., & Archer, J. A. (2023). Using foetal age estimates to substitute birth date recording in beef cattle evaluations. *Proceedings of the Association for the Advancement of Animal Breeding and Genetics*, 25, 126–129.
11. Weik, F., **Kang, J.**, & Archer, J. A. (2023). Prediction of gestation length and days to conception from foetal age scans for genetic improvement of maternal fertility in beef cow herds: a simulation study. *Proceedings of the Association for the Advancement of Animal Breeding and Genetics*, 25, 130–133.
12. Kong, Q., Mulvihill, A., Slaughter, V., Fraser, H., Cavanagh-Welch, B., Elwina, F. C., **Kang, J.**, & Ruffman, T. (2023). Not just quantity but also quality of language: Cross-cultural comparisons of maternal mental state talk in New Zealand, Australia, and China. *PLOS ONE*, 18(3), e0282480.
13. Leyva-Pérez, M. D. L. O., Vexler, L., Byrne, S., Clot, C. R., Meade, F., Griffin, D., ... **Kang, J.**, & Milbourne, D. (2022). PotatoMASH: A low-cost, genome-scanning marker system for use in potato genomics and genetics applications. *Agronomy*, 12(10), 2461.
14. **Kang, J.**, Dodds, K., Byrne, S., Faville, M., Black, M., Hess, A., ... & Brauning, R. (2021). snpGBS: A simple and flexible bioinformatics workflow to identify SNPs from genotyping-by-sequencing data. *Exploiting genetic diversity of forages to fulfil their economic and environmental roles*, 67.
15. Ruffman, T., Ruffman, C., Hill, S., Turunc, G., Park, N., Du, K., Hayhurst, J., **Kang, J.**, ... & Hunter, J. A. (2020). RWAc and SDOc: The measurement of right-wing authoritarianism and social dominance orientation in childhood. *Social Development*, 29(4), 1194–1214.
16. Riaz, A., KockAppelgren, P., Hehir, J. G., **Kang, J.**, Meade, F., Cockram, J., ... & Byrne, S. (2020). Genetic analysis using a multi-parent wheat population identifies novel sources of septoria tritici blotch resistance. *Genes*, 11(8), 887.

OPEN-SOURCE SOFTWARE

- **SimGBS.jl** (author): A Julia package for simulating large-scale genotyping-by-sequencing data. Code: [GitHub](#)
- **integrity** (author): A CRAN R package implementing the IPD Integrity Tool for reproducible assessment of clinical trial data integrity and trustworthiness. Package: [CRAN](#)
- **marp** (author): An R package for model-averaged renewal process modelling. Code: [GitHub](#)
- **snpGBS** (author): A bioinformatics workflow for identifying SNPs from genotyping-by-sequencing data. Code: [GitHub](#)

HONOURS AND RECOGNITION

2023	Third place, Emerging Geneticists Challenge, Association for the Advancement of Animal Breeding and Genetics.
2021	Finalist, Walsh Scholars of the Year, Teagasc.
2020	Harold Silverstone Memorial Prize in Statistics, University of Otago.

ACADEMIC SERVICE AND LEADERSHIP

2021–present	Executive Committee Member, New Zealand Statistical Association; supported national initiatives in statistical community engagement and early-career development.
2024–2025	Council Member, Royal Society Te Apārangi Early Career Researcher Committee; represented early-career researcher perspectives in national research-sector discussions.
2022–2024	Co-leader, Virtual Institute of Statistical Genetics, Genetics Otago; helped coordinate seminars, networking and interdisciplinary engagement across statistical genetics, genomics and quantitative genetics.
2023	Organising Committee Member, MapNet.
2022	Organising Committee Member, Early Career and Student Statisticians Mini-conference.
2021	Organising Committee Member, Genetics Otago Quantitative Genetics Symposium.